

# **ECx00K&ECx00M&ECx00N&EG810M Series**

## **FTP(S) Application Note**

**LTE Standard Module Series**

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# About the Document

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# 1 Introduction

Quectel ECx00K family, ECx00M family, ECx00N family and EG810M-CN modules support FTP and FTPS file transfer protocols (hereinafter referred to as "FTP(S)").

The FTP (File Transfer Protocol) is a standard network protocol used for the transfer of computer files between a client and server on a computer network, with high transmission rate.

FTPS (also known FTP-SSL, and FTP Secure) is an extension to the commonly used File Transfer Protocol (FTP) that adds support for the Transport Layer Security (TLS) and, formerly, the Secure Sockets Layer cryptographic protocols.

This document introduces how to use the FTP(S) function of the following Quectel modules through AT commands.

## 1.1. Applicable Modules

**Table 1: Applicable Modules**

| Module Family | Module    |
|---------------|-----------|
| ECx00M        | EC200M-CN |
|               | EC600M-CN |
|               | EC800M-CN |
| ECx00N        | EC200N-CN |
|               | EC800N-CN |
| ECx00K        | EC600K-CN |
|               | EC800K-CN |
| -             | EG810M-CN |

**NOTE**

Limited by the flash size of the module, different models of the modules support MQTT feature differently. For more details, please contact Quectel Technical Support.

## 1.2. AT Command Introduction

### 1.2.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

### 1.2.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

**Table 2: Types of AT Commands**

| Command Type      | Syntax                                                          | Description                                                                                                              |
|-------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Test Command      | <b>AT+&lt;cmd&gt;=?</b>                                         | Test the existence of the corresponding command and return information about the type, value, or range of its parameter. |
| Read Command      | <b>AT+&lt;cmd&gt;?</b>                                          | Check the current parameter value of the corresponding command.                                                          |
| Write Command     | <b>AT+&lt;cmd&gt;=&lt;p1&gt;[,&lt;p2&gt;[,&lt;p3&gt;[...]]]</b> | Set user-definable parameter value.                                                                                      |
| Execution Command | <b>AT+&lt;cmd&gt;</b>                                           | Return a specific information parameter or perform a specific action.                                                    |

### 1.3. Declaration of AT Command Examples

The AT command examples in this document are provided to help you familiarize with AT commands and learn how to use them. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples and that they should be executed in a given sequence.

### 1.4. The Process of Using FTP(S) AT Commands

As ECx00K family, ECx00M family, ECx00N family and EG810M-CN modules support FTP(S) protocol, the file and directory on FTP(S) server can be operated via FTP(S) AT commands. The general process is as follows:

#### Step 1: Configure and activate a PDP context.

- 1) Configure <APN>, <username>, <password> and other parameters of a PDP context with **AT+QICSGP**. See **document [1]** for details. If QoS settings need to be updated, configure them with **AT+CGQMIN**, **AT+CGEQMIN**, **AT+CGQREQ** and **AT+CGEQREQ** commands. See **document [2]** for details.
- 2) Activate the PDP context via **AT+QIACT**. See **document [1]** for details.
- 3) Configure the PDP context ID for FTP(S) with **AT+QFTPCFG="contextid",<contextid>**. Activate the PDP context first.

#### Step 2: Configure user account and FTP(S) server.

- 1) Configure account information with **AT+QFTPCFG="account",<username>,<password>**.
- 2) Configure of transferred data with **AT+QFTPCFG="filetype",<file\_type>**. Either binary data or ASCII data can be transferred between FTP(S) server and client.
- 3) Configure the transfer mode with **AT+QFTPCFG="transmode",<transmode>**. The transfer mode means either the FTP(S) server or client listens on a port for a data connection. **AT+QFTPCFG="transmode",1** must be set for FTPS operations, because FTPS currently does not support active mode.
- 4) Configure the response timeout value with **AT+QFTPCFG="rsptimeout",<timeout>**.
- 5) If the module is an FTPS client, execute the following commands:
  - a) Execute **AT+QFTPCFG="ssltype",1**.
  - b) Execute **AT+QFTPCFG="SSL\_ctxID",<SSL\_ctxID>** to specify <SSL\_ctxID>.
  - c) Execute **AT+QSSLCFG** to configure the specified <SSL\_ctxID>. For more details, see **document [3]**.

#### Step 3: Login to FTP(S) server.

Login to FTP(S) server with **AT+QFTPOPEN=<hostname>,<port>**. If **+QFTPOPEN: 0,0** URC is returned, it indicates a successful operation. Port numbers of FTPS and FTP servers are different. The port number of FTPS server depends on FTPS server provider, and it is usually 990.

#### Step 4: File operation.

- 1) Set the current directory with **AT+QFTPCWD**.
- 2) Upload a file to FTP(S) server.
  - a) Upload a file to UFS or SD card via **AT+QFUPL**, then upload the file to FTP(S) server with **AT+QFTPPUT**. After uploading the file to FTP(S) server successfully, delete the file with **AT+QFDEL**. For more details, see *document [4]*.
  - b) Upload a file to FTP(S) server through COM port with **AT+QFTPPUT**, then the module enters data mode. **+++** can be inputted to finish the file uploading process.
- 3) Download a file from FTP(S) server with **AT+QFTPGET**. The file can be outputted to COM port or saved to UFS or SD card. If the file is outputted to COM port, the module enters data mode.
- 4) Get the size of the file on FTP(S) server with **AT+QFTPSIZE**.
- 5) Get the length of data transferred between FTP(S) server and client with **AT+QFTPLEN**.
- 6) Delete a file on FTP(S) server with **AT+QFTPDEL**.
- 7) Rename a file on FTP(S) server with **AT+QFTPRENAME**.

#### Step 5: Directory operation on FTP(S) server.

- 1) Set the current directory with **AT+QFTPCWD**.
- 2) Create a folder with **AT+QFTPMKDIR**.
- 3) List the content of a directory with **AT+QFTPLIST**.
- 4) List file names of a directory with **AT+QFTPNLST**.
- 5) Rename a file or folder with **AT+QFTPRENAME**.
- 6) Delete a folder with **AT+QFTPRMDIR**.
- 7) List standardized file and directory information with **AT+QFTPMLSD**.
- 8) Get the file modification time on FTP(S) server with **AT+QFTPMDTM**.

#### Step 6: Close connection with FTP(S) server.

Close the connection with FTP(S) server with **AT+QFTPCLOSE**. If **+QFTPCLOSE: 0,0** URC is reported, it indicates that the operation is successful and **Step 3** to **Step 6** can be repeated.

#### Step 7: Deactivate PDP context.

Deactivate the PDP context with **AT+QIDEACT=<contextID>**. For more details, see *document [1]*.

## 1.5. Description of Data Mode

The COM port of ECx00K family, ECx00M family, ECx00N family and EG810M-CN modules has two working modes: AT command mode and data mode. In AT command mode, the data inputted via COM port are regarded as AT command, while in data mode, they are regarded as data.

Inputting **+++** or pulling up DTR (AT&D1 should be set first) will make the module exit data mode. To prevent the **+++** from being misinterpreted as data, the following sequence should be followed:

- 1) Do not input any character for at least 1 s before inputting **+++**.

- 2) Input **+++** within 1 s, making sure that no other characters are inputted during the time.
- 3) Do not input any character within 1 s after inputting **+++**.

Once **AT+QFTPPUT**, **AT+QFTPGET**, **AT+QFTPLIST** and **AT+QFTPNLST** are executed, if the local file path is "**COM:**", which means that data will be received from or outputted to COM port, the COM port enters data mode. The port exits data mode after inputting **+++** or changing DTR level from low to high, and it reenters data mode by executing **ATO** command after **AT+QFTPGET**, **AT+QFTPLIST** and **AT+QFTPNLST**. After **AT+QFTPPUT** is executed, the port cannot reenter data mode via **ATO**.

## 2 Description of FTP(S) AT Commands

### 2.1. AT+QFTPCFG Configure Parameters for FTP(S) Server

This command configures FTP(S) server parameters, including user account, file type, transfer mode and context ID. If the Write Command only executes one parameter, it will query the current settings.

| AT+QFTPCFG Configure Parameters for FTP(S) Server                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPCFG=?</b>                                              | Response<br><b>+QFTPCFG: "account",&lt;username&gt;,&lt;password&gt;</b><br><b>+QFTPCFG: "filetype",(list of supported &lt;file_type&gt;s)</b><br><b>+QFTPCFG: "transmode",(list of supported &lt;transmode&gt;s)</b><br><b>+QFTPCFG: "contextid",(range of supported &lt;contextID&gt;s)</b><br><b>+QFTPCFG: "rsptimeout",(range of supported &lt;timeout&gt;s)</b><br><b>+QFTPCFG: "ssltype",(range of supported &lt;SSL_type&gt;s)</b><br><b>+QFTPCFG: "SSL_ctxID",(range of supported &lt;SSL_ctxID&gt;s)</b><br><b>+QFTPCFG: "data_address",(range of supported &lt;data_address_type&gt;s),(range of supported &lt;data_timeout&gt;s)</b><br><b>+QFTPCFG: "restenable",(range of supported &lt;rest_enable&gt;s)</b><br><b>+QFTPCFG: "cache",(range of supported &lt;cache_type&gt;s)</b><br><br><b>OK</b> |
| Write Command<br><b>AT+QFTPCFG="account"[,&lt;username&gt;,&lt;password&gt;]</b> | Response<br>If the optional parameters are omitted, query the current configuration:<br><b>+QFTPCFG: "account",&lt;username&gt;,&lt;password&gt;</b><br><br><b>OK</b><br><br>If the optional parameters are specified, configure the user account:<br><b>OK</b><br>Or<br><b>ERROR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                               |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Write Command</p> <p><b>AT+QFTPCFG="filetype",&lt;file_type&gt;</b><br/><b>]</b></p>       | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:<br/><b>+QFTPCFG: "filetype",&lt;file_type&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure the file type:<br/><b>OK</b><br/>Or<br/><b>ERROR</b></p>                                                             |
| <p>Write Command</p> <p><b>AT+QFTPCFG="transmode",&lt;transmode&gt;</b><br/><b>]</b></p>      | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:<br/><b>+QFTPCFG: "transmode",&lt;transmode&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, set it to the FTP(S) server or a client that listens on a port for data connection:<br/><b>OK</b><br/>Or<br/><b>ERROR</b></p> |
| <p>Write Command</p> <p><b>AT+QFTPCFG="contextid",&lt;contextid&gt;</b><br/><b>D&gt;]</b></p> | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:<br/><b>+QFTPCFG: "contextid",&lt;contextID&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure the context ID:<br/><b>OK</b><br/>Or<br/><b>ERROR</b><br/>Or<br/><b>+CME ERROR: &lt;err&gt;</b></p>                 |
| <p>Write Command</p> <p><b>AT+QFTPCFG="rsptimeout",&lt;timeout&gt;</b><br/><b>t&gt;]</b></p>  | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:<br/><b>+QFTPCFG: "rsptimeout",&lt;timeout&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure the response</p>                                                                                                     |

|                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                               | <p>timeout:</p> <p><b>OK</b></p> <p>Or</p> <p><b>ERROR</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>Write Command</p> <p><b>AT+QFTPCFG="ssltype"[,&lt;SSL_type&gt;]</b></p>                                    | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:</p> <p><b>+QFTPCFG: "ssltype",&lt;SSL_type&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure the SSL type:</p> <p><b>OK</b></p> <p>Or</p> <p><b>ERROR</b></p>                                                                                                                                                                                                                       |
| <p>Write Command</p> <p><b>AT+QFTPCFG="SSL_ctxID"[,&lt;SSL_ctxID&gt;]</b></p>                                 | <p>Response</p> <p>If the optional parameter is omitted, query the current configuration:</p> <p><b>+QFTPCFG: "SSL_ctxID",&lt;SSL_ctxID&gt;</b></p> <p><b>OK</b></p> <p>If the optional parameter is specified, configure the SSL context ID:</p> <p><b>OK</b></p> <p>Or</p> <p><b>ERROR</b></p>                                                                                                                                                                                                              |
| <p>Write Command</p> <p><b>AT+QFTPCFG="data_address"[,&lt;data_address_type&gt;,&lt;data_timeout&gt;]</b></p> | <p>Response</p> <p>If the optional parameters are omitted, query the current configuration:</p> <p>1) &lt;data_address_type&gt; equals 2:</p> <p><b>+QFTPCFG: "data_address",&lt;data_address_type&gt;,&lt;data_timeout&gt;</b></p> <p><b>OK</b></p> <p>2) &lt;data_address_type&gt; is not equal to 2:</p> <p><b>+QFTPCFG: "data_address",&lt;data_address_type&gt;</b></p> <p><b>OK</b></p> <p>If any of the optional parameters are specified, configure the data connection address:</p> <p><b>OK</b></p> |

|                                                                      |                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | Or<br><b>ERROR</b>                                                                                                                                                                                                                                                                  |
| Write Command<br><b>AT+QFTPCFG="restenable",&lt;rest_enable&gt;]</b> | Response<br>If the optional parameter is omitted, query the current configuration:<br><b>+QFTPCFG: "restenable",&lt;rest_enable&gt;</b><br><br><b>OK</b><br><br>If the optional parameter is specified, configure the way to close data session:<br><b>OK</b><br>Or<br><b>ERROR</b> |
| Write Command<br><b>AT+QFTPCFG="cache",&lt;cache_type&gt;]</b>       | Response<br>If the optional parameter is omitted, query the current configuration:<br><b>+QFTPCFG: "cache",&lt;cache_type&gt;</b><br><br><b>OK</b><br><br>If the optional parameter is specified, configure the size and type of WM:<br><b>OK</b><br>Or<br><b>ERROR</b>             |
| Maximum Response Time                                                | 300 ms                                                                                                                                                                                                                                                                              |
| Characteristics                                                      | This command takes effect immediately.<br>The configuration is not saved.                                                                                                                                                                                                           |

## Parameter

|                          |                                                                                                                                                                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;username&gt;</b>  | String type. Username for authentication. Maximum size: 255 bytes.                                                                                                                                                                                                                                    |
| <b>&lt;password&gt;</b>  | String type. The password for authentication. Maximum size: 255 bytes.                                                                                                                                                                                                                                |
| <b>&lt;file_type&gt;</b> | Integer type. Type of transferred data.<br><u>0</u> Binary<br>1 ASCII                                                                                                                                                                                                                                 |
| <b>&lt;transmode&gt;</b> | Integer type. Determines whether the FTP(S) server or client listens on a port for data connection.<br>0 Active mode, the module will listen on a port for a data connection from the FTP server<br><u>1</u> Passive mode, the FTP server will listen on a port for a data connection from the client |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;contextID&gt;</b>         | Integer type. PDP context ID. Range: 1–15. Default value: 1. Activate it with <b>AT+QIACT</b> before using <b>AT+QFTPOPEN</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>&lt;timeout&gt;</b>           | Integer type. Range: 20–180. Default value: 90. Unit: second.<br>Generally, it is the maximum response time for most <b>+QFTPXXX: xx,xx</b> commands after the <b>OK</b> result code is returned, except <b>AT+QFTPPUT/QFTPGET/QFTPLST/QFTPNLST</b> commands. The rules for these four commands are shown below:<br>a) This parameter indicates the maximum response time for <b>CONNECT</b> to be outputted after the command has been sent.<br>b) When the module has entered data mode, this parameter indicates the maximum interval between two packets of received/transferred data.<br>c) When the <b>&lt;local_name&gt;</b> is not " <b>COM:</b> ", it indicates the maximum interval between two packets of received/transferred data. |
| <b>&lt;SSL_type&gt;</b>          | Integer type. The module works as an FTP client or FTPS client.<br>0 FTP client<br>1 FTPS implicit encryption<br>2 FTPS explicit encryption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>&lt;SSL_ctxID&gt;</b>         | Integer type. SSL context ID. Range: 0–5. Default value: 0. You can configure the SSL parameters with <b>AT+QSSLCFG</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>&lt;data_address_type&gt;</b> | Integer type. Determines the FTP(S) data connection address.<br>0 Use server dispatched address<br>1 Use FTP(S) control session address<br>2 Use FTP(S) control session address in priority. If the connection fails, an address assigned by server will be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>&lt;data_timeout&gt;</b>      | Integer type. The time required to switch the address assigned by the server when the FTP(S) control session address connection fails. Valid only when <b>&lt;data_address_type&gt;</b> is 2. Range: 15–50. Default value: 25. Unit: s.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>&lt;rest_enable&gt;</b>       | Integer type. The way to close data session.<br>0 Wavehand<br>1 Sending RST package                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>&lt;cache_type&gt;</b>        | Integer type. The size and type of WM.<br>0 Default mode. Cache size: 64 KB.<br>1 Quick mode. Cache size: 128 KB.<br>2 Slow mode. Cache size: 128 KB. Resume the download.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>&lt;err&gt;</b>               | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**NOTE**

Since FTPS currently does not support active mode, **<transmode>** must be set to 1 during FTPS operation.

## 2.2. AT+QFTPOPEN Login to FTP(S) Server

This command logs in to FTP(S) server. Activate the PDP context with **AT+QIACT** first. **+QFTPOPEN: <err>,<protocol\_error>** indicates the operation result of **AT+QFTPOPEN** and it should be outputted within **<timeout>** configured with **AT+QFTPCFG**.

| AT+QFTPOPEN Login to FTP(S) Server                                  |                                                                                                                           |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPOPEN=?</b>                                | Response<br><b>+QFTPOPEN: &lt;hostname&gt;,&lt;port&gt;</b><br><br><b>OK</b>                                              |
| Write Command<br><b>AT+QFTPOPEN=&lt;hostname&gt;[,&lt;port&gt;]</b> | Response<br><b>OK</b><br><br><b>+QFTPOPEN: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                               | 125 s                                                                                                                     |
| Characteristics                                                     | This command takes effect immediately.<br>The configuration is not saved.                                                 |

### Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;hostname&gt;</b>       | String type. IP address or domain name of the FTP(S) server.<br>Maximum parameter size: 255 bytes.                                                                                     |
| <b>&lt;port&gt;</b>           | Integer type. FTP(S) server port number. Default value: 21. Range: 1–65535.                                                                                                            |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

### Example

```

AT+QFTPOPEN="101.95.183.122",21 //Login to FTP server.
OK

+QFTPOPEN: 0,0
AT+QFTPOPEN="quectel.3322.org",990 //Login to FTPS server.
OK

+QFTPOPEN: 0,0

```

**NOTE**

Note that the FTPS and FTP server port numbers are different. The FTPS server port number depends on FTPS server provider, and it is usually 990.

## 2.3. AT+QFTPCWD Configure the Current Directory on FTP(S) Server

The command configures the current directory on FTP(S) server. If **OK** is returned, **+QFTPCWD: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. All the file and directory operations will be configured in the current directory.

| AT+QFTPCWD Configure the Current Directory on FTP(S) Server |                                                                                                                          |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPCWD=?</b>                         | Response<br><b>+QFTPCWD: &lt;path_name&gt;</b><br><br><b>OK</b>                                                          |
| Write Command<br><b>AT+QFTPCWD=&lt;path_name&gt;</b>        | Response<br><b>OK</b><br><br><b>+QFTPCWD: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                       | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                   |
| Characteristics                                             | This command takes effect immediately.<br>The configuration is not saved.                                                |

### Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;path_name&gt;</b>      | String type. A directory path on FTP(S) server. Maximum size: 128 bytes. FTP(S) server root path: "/".                                                                                 |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.4. AT+QFTPPWD Get the Current Directory on FTP(S) Server

This command gets the current directory on FTP(S) server. If **OK** is returned, **+QFTPPWD: 0,<path\_name>** or **+QFTPPWD: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**.

| AT+QFTPPWD Get the Current Directory on FTP(S) Server |                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPPWD=?</b>                   | Response<br><b>OK</b>                                                                                                                                                                                                                                                 |
| Execution Command<br><b>AT+QFTPPWD</b>                | Response<br><b>OK</b><br><br>If the current directory is gotten successfully:<br><b>+QFTPPWD: 0,&lt;path_name&gt;</b><br><br>If it fails to get the current directory:<br><b>+QFTPPWD: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                 | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                |
| Characteristics                                       | This command takes effect immediately.<br>The configuration is not saved.                                                                                                                                                                                             |

### Parameter

|                               |                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;path_name&gt;</b>      | String type. Directory path on FTP(S) server. Maximum size: 128 bytes. FTP(S) server root path: "/".                                                                                      |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                    |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> .<br>If it is 0, it is invalid. |

## 2.5. AT+QFTPPUT Upload a File to FTP(S) Server

This command uploads a file to FTP(S) server. The file data can be uploaded via COM port, and then the module will enter data mode. Inputting **+++** can abort the file uploading. A local file can be uploaded to FTP(S) server and the file can be a UFS or an SD file. The file can be uploaded to UFS or SD card with **AT+QFUPL**, and then uploaded to FTP(S) server via **AT+QFTPPUT** command. After the file is uploaded successfully, it can be deleted with **AT+QFDEL**. For more details, see **document [4]**.

A file can be uploaded from a specified file position with **<startpos>**. If the **<local\_name>** is "COM:", **CONNECT** should be outputted within **<timeout>** configured with **+QFTPPUT: <err>,<protocol\_error>**. If the **<local\_name>** is not "COM:", **OK** is outputted first, and then **+QFTPPUT: 0,<transferlen>** is outputted after completing the data transfer.

If the module has entered data mode or the **<local\_name>** is not "COM:", the **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transferred data.

| AT+QFTPPUT Upload a File to FTP(S) Server                                                                                                                             |                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPPUT=?</b>                                                                                                                                   | Response<br><b>+QFTPPUT: &lt;file_name&gt;,&lt;local_name&gt;,&lt;startpos&gt;,&lt;uploadlen&gt;,&lt;beof&gt;</b><br><br><b>OK</b>                                                                                                                                                        |
| Write Command<br>When <b>&lt;local_name&gt;</b> is "COM:"<br><b>AT+QFTPPUT=&lt;file_name&gt;,&lt;local_name&gt;,&lt;startpos&gt;,&lt;uploadlen&gt;,&lt;beof&gt;]]</b> | Response<br><b>CONNECT</b><br><Input file data><br><b>OK</b><br><br>If the file is uploaded successfully:<br><b>+QFTPPUT: 0,&lt;transferlen&gt;</b><br><br>If it fails to upload the file:<br><b>+QFTPPUT: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Write Command<br>When <b>&lt;local_name&gt;</b> is not "COM:"<br><b>AT+QFTPPUT=&lt;file_name&gt;,&lt;local_name&gt;,&lt;startpos&gt;]</b>                             | Response<br><b>OK</b><br><br>If the file is uploaded successfully:<br><b>+QFTPPUT: 0,&lt;transferlen&gt;</b><br><br>If it fails to upload the file:<br><b>+QFTPPUT: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b>                                        |
| Maximum Response Time                                                                                                                                                 | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                                    |
| Characteristics                                                                                                                                                       | This command takes effect immediately.<br>The configuration is not saved.                                                                                                                                                                                                                 |

## Parameter

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <file_name>      | String type. File name on FTP(S) server. Maximum size: 255 bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <local_name>     | String type. Local file name. Maximum size: 255 bytes. If it is " <b>COM:</b> ", the data are inputted via COM port. If it is not " <b>COM:</b> ", the data are saved to UFS or SD card. After successful uploading, the file can be deleted with <b>AT+QFDEL</b> .                                                                                                                                                                                                                                                                                                                                                                     |
| <startpos>       | Integer type. Start position of the file to be uploaded. Default value: 0. If <uploadlen> and <beof> are specified, <startpos> should be the position where the data continue to be uploaded to the same file. Unit: byte.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <uploadlen>      | Integer type. Length of the data to be uploaded. It is valid only when <local_name> is " <b>COM:</b> ". When the length of the data uploaded via COM port reaches <uploadlen>, the module exits data mode. Unit: byte.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <beof>           | Integer type. Determines whether it is the last data packet to be uploaded. It is valid only when <local_name> is " <b>COM:</b> ".<br>0 Not the last data packet. When the data length reaches <uploadlen>, the module exits data mode, and <b>+QFTPPUT: 0,&lt;transferlen&gt;</b> is outputted. In such a case, do not disconnect data connection, as the remaining data need to be uploaded to the same file on FTP(S)<br><u>1</u> The last data packet. When the data length reaches <uploadlen>, the module exits data mode and data connection can be disconnected. After that <b>+QFTPPUT: 0,&lt;transferlen&gt;</b> is outputted |
| <transferlen>    | Integer type. Length of successfully transferred data. Unit: byte.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <err>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <protocol_error> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### NOTE

If <uploadlen> and <beof> are not specified, all data will be uploaded to FTP(S) server.

## Example

//Upload a file via COM port to FTP(S) server.

**AT+QFTPPUT="test.txt","COM:",0**

//All data will be saved as *test.txt* on FTP(S) server.

**CONNECT**

<Input file data>

<++++>

**OK**

**+QFTPPUT: 0,1000**

//Upload a file via COM port to FTP(S) server twice in 1024-byte chunks.

```

AT+QFTPPUT="test.txt","COM:",0,1024,0           //It is not the last 1024-byte chunk of test.txt.
CONNECT
<Input file data>
OK                                               //Data length reaches 1024 bytes.

+QFTPPUT: 0,1024
AT+QFTPPUT="test.txt","COM:",1024,1024,1       //It is the last 1024-byte chunk of test.txt.
CONNECT
<Input file data>
OK                                               //Data length reaches 1024 bytes.

+QFTPPUT: 0,1024

```

## 2.6. AT+QFTPGET Download a File from FTP(S) Server

This command downloads a file from FTP(S) server. The file can be outputted via COM port with **AT+QFTPGET="filename","COM:"**. The module enters data mode on receiving data from server. After the data transfer is completed, the module exits data mode automatically and outputs **QFTPGET: 0,<transferlen>**. The file can be saved to UFS or SD card with **AT+QFTPGET="filename","UFS:localname"** or **AT+QFTPGET="filename","SD:localname"**. After the file transfer is completed, the module outputs **+QFTPGET: 0,<transferlen>**.

If the **<local\_name>** is "COM:", **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If the **<local\_name>** is not "COM:", **OK** is outputted first, and then **+QFTPGET: 0,<transferlen>** after the data transfer is completed.

If the module has entered data mode or the **<local\_name>** is not "COM:", the **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transferred data.

### AT+QFTPGET Download a File from FTP(S) Server

|                                                                                                                                                           |                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPGET=?</b>                                                                                                                       | Response<br><b>+QFTPGET:</b><br><b>&lt;file_name&gt;,&lt;local_name&gt;,&lt;startpos&gt;,&lt;downloadlen&gt;</b><br><br><b>OK</b>                                   |
| Write Command<br>When <b>&lt;local_name&gt;</b> is "COM:"<br><b>AT+QFTPGET=&lt;file_name&gt;,&lt;local_name&gt;,&lt;startpos&gt;,&lt;downloadlen&gt;]</b> | Response<br><b>CONNECT</b><br><b>&lt;Output file data&gt;</b><br><b>OK</b><br><br>If the file is downloaded successfully:<br><b>+QFTPGET: 0,&lt;transferlen&gt;</b> |

|                                                                                                                                                       |                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                       | <p>If it fails to download the file:<br/> <b>+QFTPGET: &lt;err&gt;,&lt;protocol_error&gt;</b><br/> Or<br/> <b>+CME ERROR: &lt;err&gt;</b></p>                                                                                                                           |
| <p>Write Command<br/> When <b>&lt;local_name&gt;</b> is not "COM:"<br/> <b>AT+QFTPGET=&lt;file_name&gt;,&lt;local_name&gt;[,&lt;startpos&gt;]</b></p> | <p>Response<br/> <b>OK</b></p> <p>If the file is downloaded successfully:<br/> <b>+QFTPGET: 0,&lt;transferlen&gt;</b></p> <p>If it fails to download the file:<br/> <b>+QFTPGET: &lt;err&gt;,&lt;protocol_error&gt;</b><br/> Or<br/> <b>+CME ERROR: &lt;err&gt;</b></p> |
| Maximum Response Time                                                                                                                                 | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                  |
| Characteristics                                                                                                                                       | <p>This command takes effect immediately.</p> <p>The configuration is not saved.</p>                                                                                                                                                                                    |

## Parameter

|                               |                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;file_name&gt;</b>      | String type. File name on FTP(S) server. Maximum size: 255 bytes.                                                                                                                                                                                                                                                                                                    |
| <b>&lt;local_name&gt;</b>     | String type. Local file name. Maximum size: 255 bytes. If it is "COM:", the file data are outputted via COM port. If it is not "COM:", the data are saved to UFS or SD card. It is strongly recommended to save the file in UFS or SD card. Then the file can be read with <b>AT+QFREAD</b> . For more details, see <i>document [4]</i> .                            |
| <b>&lt;startpos&gt;</b>       | Integer type. Start position of the file to be downloaded. Default value: 0. Unit: byte.                                                                                                                                                                                                                                                                             |
| <b>&lt;downloadlen&gt;</b>    | Integer type. Length of data to be downloaded. It is valid only when <b>&lt;local_name&gt;</b> is "COM:". If this parameter is specified, the module will output <b>&lt;downloadlen&gt;</b> bytes to COM port and exit data mode. The remaining data, if any, can be downloaded from <b>&lt;startpos&gt;</b> with the same AT command. Default value: 0. Unit: byte. |
| <b>&lt;transferlen&gt;</b>    | Integer type. Length of actually transferred data. If it is less than <b>&lt;downloadlen&gt;</b> , it means the whole file has been transferred. Unit: byte.                                                                                                                                                                                                         |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                                                                                                                                               |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid.                                                                                                                                                                               |

### NOTE

When **<local\_name>** is "COM:", if **<startpos>** and **<downloadlen>** are not specified, a whole file will be downloaded from FTP(S) server.

## Example

```
//Download a file. The file is outputted via COM port.

AT+QFTPGET="test.txt","COM:",0           //Download a whole file from FTP(S) server.
CONNECT
<Output file data>
OK

+QFTPGET: 0,1000

//Download a file. The file is outputted via COM port twice in 500-byte chunks.

AT+QFTPGET="test.txt","COM:",0,500       //The size of test.txt is 1000 bytes. Download the first 500
                                           bytes.
CONNECT
<Output file data>
OK

+QFTPGET: 0,500
AT+QFTPGET="test.txt","COM:",500,500     //Download the remaining 500 bytes.
CONNECT
<Output file data>
OK

+QFTPGET: 0,500
```

## 2.7. AT+QFTPSIZE Get File Size on FTP(S) Server

This command gets the file size on FTP(S) server. If **OK** is returned, **+QFTPSIZE: 0,<file\_size>** or **+QFTPSIZE: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and reactivated.

| AT+QFTPSIZE Get File Size on FTP(S) Server            |                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPSIZE=?</b>                  | Response<br><b>+QFTPSIZE: &lt;file_name&gt;</b><br><br><b>OK</b>      |
| Write Command<br><b>AT+QFTPSIZE=&lt;file_name&gt;</b> | Response<br><b>OK</b><br><br>If the file size is gotten successfully: |

|                       |                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <b>+QFTPSIZE: 0,&lt;file_size&gt;</b><br><br>If it fails to get the file size:<br><b>+QFTPSIZE: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                         |
| Characteristics       | This command takes effect immediately.<br>The configuration is not saved.                                                                                                      |

## Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;file_name&gt;</b>      | String type. File name on FTP(S) server. Maximum size: 255 bytes.                                                                                                                      |
| <b>&lt;file_size&gt;</b>      | Integer type. File size on FTP(S) server. Unit: byte.                                                                                                                                  |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.8. AT+QFTPDEL Delete a File on FTP(S) Server

This command deletes a specified file on FTP(S) server. If **OK** is returned, **+QFTPDEL: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and reactivated.

| <b>AT+QFTPDEL Delete a File on FTP(S) Server</b>     |                                                                                                                          |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPDEL=?</b>                  | Response<br><b>+QFTPDEL: &lt;file_name&gt;</b><br><br><b>OK</b>                                                          |
| Write Command<br><b>AT+QFTPDEL=&lt;file_name&gt;</b> | Response<br><b>OK</b><br><br><b>+QFTPDEL: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                   |
| Characteristics                                      | /                                                                                                                        |

## Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;file_name&gt;</b>      | String type. File name FTP(S) server. Maximum size: 255 bytes.                                                                                                                         |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.9. AT+QFTPMKDIR Create a Folder on FTP(S) Server

This command creates a folder on FTP(S) server. If **OK** is returned, **+QFTPMKDIR: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and reactivated.

| AT+QFTPMKDIR Create a Folder on FTP(S) Server            |                                                                                                                            |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPMKDIR=?</b>                    | Response<br><b>+QFTPMKDIR: &lt;folder_name&gt;</b><br><br><b>OK</b>                                                        |
| Write Command<br><b>AT+QFTPMKDIR=&lt;folder_name&gt;</b> | Response<br><b>OK</b><br><br><b>+QFTPMKDIR: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                    | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                     |
| Characteristics                                          | /                                                                                                                          |

## Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;folder_name&gt;</b>    | String type. Folder name on FTP(S) server. Maximum size: 255 bytes.                                                                                                                    |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.10. AT+QFTPRMDIR Delete a Folder on FTP(S) Server

This command deletes a specified folder on FTP(S) server. If **OK** is returned, **+QFTPRMDIR: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected and the network should be deactivated and reactivated.

| AT+QFTPRMDIR Delete a Folder on FTP(S) Server            |                                                                                                                            |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPRMDIR=?</b>                    | Response<br><b>+QFTPRMDIR: &lt;folder_name&gt;</b><br><br><b>OK</b>                                                        |
| Write Command<br><b>AT+QFTPRMDIR=&lt;folder_name&gt;</b> | Response<br><b>OK</b><br><br><b>+QFTPRMDIR: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                                    | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                     |
| Characteristics                                          | /                                                                                                                          |

### Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;folder_name&gt;</b>    | String type. Folder name on FTP(S) server. Maximum size: 255 bytes.                                                                                                                    |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.11. AT+QFTPLIST List Content of a Directory on FTP(S) Server

This command lists the content of a directory on FTP(S) server. If the **<local\_name>** is **"COM:"**, **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If the **<local\_name>** is not **"COM:"**, **OK** is returned first. And then **+QFTPLIST: 0,<transfer\_size>** is outputted after the content has been transferred completely.

If the module has entered data mode or the **<local\_name>** is not **"COM:"**, the **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transferred data.

| AT+QFTPLIST List Content of a Directory on FTP(S) Server                                                               |                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPLIST=?</b>                                                                                   | Response<br><b>+QFTPLIST: &lt;dirname&gt;,&lt;local_name&gt;</b><br><br><b>OK</b>                                                                                                                                                                                                                                                    |
| Write Command<br>When <b>&lt;local_name&gt;</b> is "COM:"<br><b>AT+QFTPLIST=&lt;dirname&gt;[,&lt;local_name&gt;]</b>   | Response<br><b>CONNECT</b><br><b>&lt;Output content data&gt;</b><br><b>OK</b><br><br>If the directory content is listed successfully:<br><b>+QFTPLIST: 0,&lt;transfer_size&gt;</b><br><br>If it fails to list the directory content:<br><b>+QFTPLIST: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Write Command<br>When <b>&lt;local_name&gt;</b> is not "COM:"<br><b>AT+QFTPLIST=&lt;dirname&gt;,&lt;local_name&gt;</b> | Response<br><b>OK</b><br><br>If the directory content is listed successfully:<br><b>+QFTPLIST: 0,&lt;transfer_size&gt;</b><br><br>If it fails to list the directory content:<br><b>+QFTPLIST: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b>                                                         |
| Maximum Response Time                                                                                                  | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                                                                               |
| Characteristics                                                                                                        | This command takes effect immediately.<br>The configurations are not saved.                                                                                                                                                                                                                                                          |

## Parameter

|                              |                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;dirname&gt;</b>       | String type. Folder name on FTP(S) server. Maximum size: 255 bytes. If it is ".", the command will list the content of the current directory configured with <b>AT+QFTPCWD</b> .                                                                                                                                                           |
| <b>&lt;local_name&gt;</b>    | String type. Local storage location of the data from FTP(S) server. Maximum parameter size: 60 bytes. Default value: "COM:". If it is "COM:", the data are outputted to COM port. If it is not "COM:", the data are saved to UFS or SD first. Then the file can be read via <b>AT+QFREAD</b> . For more details, see <b>document [4]</b> . |
| <b>&lt;transfer_size&gt;</b> | Integer type. Size of data transferred from FTP(S) server. Unit: byte.                                                                                                                                                                                                                                                                     |

|                  |                                                                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <err>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <protocol_error> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

### Example

```
//Get the content of current directory on FTP(S) server and the data are outputted via COM port.
```

```
AT+QFTPLIST="."
```

```
CONNECT
```

```
<Output content data>
```

```
OK
```

```
+QFTPLIST: 0,1000
```

## 2.12. AT+QFTPNLST List File Names of a Directory on FTP(S) Server

This command lists file names of a directory on FTP(S) server. If the <local\_name> is "COM:", **CONNECT** should be outputted within <timeout> configured with **AT+QFTPCFG**. If the <local\_name> is not "COM:", **OK** is returned first. And then **+QFTPNLST: 0,<transfer\_size>** after file name transferred is completed.

If the module has entered data mode or the <local\_name> is not "COM:", the <timeout> configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transferred data.

### AT+QFTPNLST List File Names of a Directory on FTP(S) Server

|                                                                                                         |                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPNLST=?</b>                                                                    | Response<br><b>+QFTPNLST: &lt;dirname&gt;,&lt;local_name&gt;</b><br><br><b>OK</b>                                                                                                                                                                                                                               |
| Write Command<br>When <local_name> is "COM:"<br><b>AT+QFTPNLST=&lt;dirname&gt;[,&lt;local_name&gt;]</b> | Response<br><b>CONNECT</b><br><b>&lt;Output content data&gt;</b><br><b>OK</b><br><br>If file names are listed successfully:<br><b>+QFTPNLST: 0,&lt;transfer_size&gt;</b><br><br>If it fails to list file names:<br><b>+QFTPNLST: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |

|                                                                                                                                       |                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Write Command</p> <p>When <b>&lt;local_name&gt;</b> is not "COM:"</p> <p><b>AT+QFTPNLST=&lt;dirname&gt;,&lt;local_name&gt;</b></p> | <p>Response</p> <p><b>OK</b></p> <p><b>+QFTPNLST: 0,&lt;transfer_size&gt;</b></p> <p>Or</p> <p><b>+CME ERROR: &lt;err&gt;</b></p> <p>If an error occurred while listing file names:</p> <p><b>+QFTPNLST: &lt;err&gt;,&lt;protocol_error&gt;</b></p> |
| Maximum Response Time                                                                                                                 | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                              |
| Characteristics                                                                                                                       | <p>This command takes effect immediately.</p> <p>The configuration is not saved.</p>                                                                                                                                                                |

## Parameter

|                               |                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;dirname&gt;</b>        | String type. Folder name on FTP(S) server. Maximum size: 255 bytes. If it is ".", it will list the file names of the current directory configured with <b>AT+QFTPCWD</b> .                                                                                                                                                                |
| <b>&lt;local_name&gt;</b>     | String type. Local storage location of the data from FTP(S) server. Maximum size: 255 bytes. Default value: "COM:". If it is "COM:", the data are outputted to COM port. If it is not "COM:", the data are saved to UFS or SD card. Then the file can be read via <b>AT+QFREAD</b> . For more details, see <a href="#">document [4]</a> . |
| <b>&lt;transfer_size&gt;</b>  | Integer type. Size of data transferred from FTP(S) server. Unit: byte.                                                                                                                                                                                                                                                                    |
| <b>&lt;err&gt;</b>            | Error code. See <a href="#">Chapter 5</a> for more information.                                                                                                                                                                                                                                                                           |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <a href="#">Chapter 6</a> . If it is 0, it is invalid.                                                                                                                                           |

## Example

```
//Get the file name of current directory on FTP(S) server and the data are outputted via COM port.
```

```
AT+QFTPNLST="."
```

```
CONNECT
```

```
<Output content data>
```

```
OK
```

```
+QFTPNLST: 0,1000
```

## 2.13. AT+QFTPMLSD List Standardized File and Directory Information

This command lists standardized file and directory information on FTP(S) server. If the **<local\_name>** is "COM:", **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If the **<local\_name>** is not "COM:", **OK** is returned first and then **+QFTPMLSD: 0,<transfer\_size>** after the content transfer is completed.

If the module has entered data mode or the **<local\_name>** is not "COM:", the **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transferred data.

| AT+QFTPMLSD List Standardized File and Directory Information                                                           |                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPMLSD=?</b>                                                                                   | Response<br><b>+QFTPMLSD: &lt;dirname&gt;,&lt;local_name&gt;</b><br><br><b>OK</b>                                                                                                                                                                                                                                                                                                        |
| Write Command<br>When <b>&lt;local_name&gt;</b> is "COM:"<br><b>AT+QFTPMLSD=&lt;dirname&gt;[,&lt;local_name&gt;]</b>   | Response<br><b>CONNECT</b><br><b>&lt;Output content data&gt;</b><br><b>OK</b><br><br>If the standardized file and directory information is listed successfully:<br><b>+QFTPMLSD: 0,&lt;transfer_size&gt;</b><br><br>If it fails to list the standardized file and directory information:<br><b>+QFTPMLSD: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Write Command<br>When <b>&lt;local_name&gt;</b> is not "COM:"<br><b>AT+QFTPMLSD=&lt;dirname&gt;,&lt;local_name&gt;</b> | Response<br><b>OK</b><br><br>If the standardized file and directory information is listed successfully:<br><b>+QFTPMLSD: 0,&lt;transfer_size&gt;</b><br><br>If it fails to list the standardized file and directory information:<br><b>+QFTPMLSD: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b>                                                         |
| Maximum Response Time                                                                                                  | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                                                                                                                                   |
| Characteristics                                                                                                        | /                                                                                                                                                                                                                                                                                                                                                                                        |

## Parameter

|                  |                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <dirname>        | String type. Folder name on FTP(S) server. Maximum size: 255 bytes. If it is ".", it will list standardized file and directory information configured with <b>AT+QFTPCWD</b> .                                                                                                                                                               |
| <local_name>     | String type. The local storage location of the data from FTP(S) server. Maximum size: 255 bytes. Default value: "COM:". If it is "COM:", the data will be outputted to COM port. If it is not "COM:", the data will be saved to UFS or SD card. Then the file can be read via <b>AT+QFREAD</b> . For more details, see <b>document [4]</b> . |
| <transfer_size>  | Integer type. Size of data transferred from FTP(S) server. Unit: byte.                                                                                                                                                                                                                                                                       |
| <err>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                                                                                                                       |
| <protocol_error> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid.                                                                                                                                                       |

## Example

```
//Get standardized file and directory information on FTP(S) server and the data are outputted via COM
port.

AT+QFTPMLSD="."
CONNECT
<Output content data>
OK

+QFTPMLSD: 0,1000
```

## 2.14. AT+QFTPMDTM Get the File Modification Time on FTP(S) Server

This command gets the file modification time on FTP(S) server. If **OK** is returned, **+QFTPMDTM: 0,<modify\_time>** or **+QFTPMDTM: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and reactivated.

| AT+QFTPMDTM Get the File Modification Time on FTP(S) Server |                                                                  |
|-------------------------------------------------------------|------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPMDTM=?</b>                        | Response<br><b>+QFTPMDTM: &lt;file_name&gt;</b><br><br><b>OK</b> |
| Write Command<br><b>AT+QFTPMDTM=&lt;file_name&gt;</b>       | Response<br><b>OK</b>                                            |

|                       |                                                                                                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>If the file modification time is gotten successfully:<br/> <b>+QFTPMDTM: 0,&lt;modify_time&gt;</b></p> <p>If it fails to get the file modification time:<br/> <b>+QFTPMDTM: &lt;err&gt;,&lt;protocol_error&gt;</b><br/> Or<br/> <b>+CME ERROR: &lt;err&gt;</b></p> |
| Maximum Response Time | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                                                                                                                                                |
| Characteristics       | /                                                                                                                                                                                                                                                                     |

## Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;file_name&gt;</b>      | String type. File name FTP(S) server. Maximum size: 255 bytes.                                                                                                                         |
| <b>&lt;modify_time&gt;</b>    | String type. File modification time on FTP(S) server.<br>Format: "YYYYMMDDHHMMSS" or "YYYYMMDDHHMMSS.NNN".                                                                             |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.15. AT+QFTPRENAME Rename a File or Folder on FTP(S) Server

This command renames a file or folder on FTP(S) server. If **OK** is returned, **+QFTPRENAME: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and reactivated.

| <b>AT+QFTPRENAME Rename a File or Folder on FTP(S) Server</b>           |                                                                                                                                          |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPRENAME=?</b>                                  | <p>Response<br/> <b>+QFTPRENAME: &lt;old_name&gt;,&lt;new_name&gt;</b></p> <p><b>OK</b></p>                                              |
| Write Command<br><b>AT+QFTPRENAME=&lt;old_name&gt;,&lt;new_name&gt;</b> | <p>Response<br/> <b>OK</b></p> <p><b>+QFTPRENAME: &lt;err&gt;,&lt;protocol_error&gt;</b><br/> Or<br/> <b>+CME ERROR: &lt;err&gt;</b></p> |
| Maximum Response Time                                                   | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                                   |
| Characteristics                                                         | /                                                                                                                                        |

## Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;old_name&gt;</b>       | String type. The old file name or folder name on FTP(S) server. Maximum size: 255 bytes.                                                                                               |
| <b>&lt;new_name&gt;</b>       | String type. The new file name or folder name on FTP(S) server. Maximum size: 255 bytes.                                                                                               |
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

## 2.16. AT+QFTPLEN Get the Length of Transferred Data

This command gets the length of transferred data on FTP(S) server.

| AT+QFTPLEN Get the Length of Transferred Data |                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPLEN=?</b>           | Response<br><b>OK</b>                                                                                       |
| Execution Command<br><b>AT+QFTPLEN</b>        | Response<br><b>OK</b><br><br><b>+QFTPLEN: 0,&lt;transferlen&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                         | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                      |
| Characteristics                               | /                                                                                                           |

## Parameter

|                            |                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;transferlen&gt;</b> | Integer type. The length of transferred data on FTP(S) server. When executing <b>AT+QFTPPUT</b> , <b>AT+QFTPGET</b> , <b>AT+QFTPNLST</b> or <b>AT+QFTPLIST</b> , the length of transferred data can be queried by <b>AT+QFTPLEN</b> . Unit: byte. |
| <b>&lt;err&gt;</b>         | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                            |

## 2.17. AT+QFTPSTAT Get the Status of FTP(S) Server

This command gets the status of FTP(S) server.

| AT+QFTPSTAT Get the Status of FTP(S) Server |                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPSTAT=?</b>        | Response<br><b>OK</b>                                                                                    |
| Execution Command<br><b>AT+QFTPSTAT</b>     | Response<br><b>OK</b><br><br><b>+QFTPSTAT: 0,&lt;ftpstat&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                       | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                   |
| Characteristics                             | /                                                                                                        |

### Parameter

|                        |                                                                                                                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;ftpstat&gt;</b> | Integer type. Current status of FTP(S) server.<br>0      Opening an FTP(S) server<br>1      FTP(S) server is opened and idle<br>2      Transferring data with FTP(S) server<br>3      Closing the FTP(S) server<br>4      FTP(S) server is closed |
| <b>&lt;err&gt;</b>     | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                                                                            |

## 2.18. AT+QFTPCLOSE Log out from FTP(S) Server

This command logs out from FTP(S) server. If **OK** is returned, **+QFTPCLOSE: <err>,<protocol\_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the network should be deactivated and reactivated.

| AT+QFTPCLOSE Log out from FTP(S) Server  |                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Test Command<br><b>AT+QFTPCLOSE=?</b>    | Response<br><b>OK</b>                                                                                                      |
| Execution Command<br><b>AT+QFTPCLOSE</b> | Response<br><b>OK</b><br><br><b>+QFTPCLOSE: &lt;err&gt;,&lt;protocol_error&gt;</b><br>Or<br><b>+CME ERROR: &lt;err&gt;</b> |
| Maximum Response Time                    | Determined by <b>&lt;timeout&gt;</b> configured with <b>AT+QFTPCFG</b>                                                     |
| Characteristics                          | This command takes effect immediately.<br>The configuration is not saved.                                                  |

### Parameter

|                               |                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;err&gt;</b>            | Error code. See <b>Chapter 5</b> for more information.                                                                                                                                 |
| <b>&lt;protocol_error&gt;</b> | Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more details, see <b>Chapter 6</b> . If it is 0, it is invalid. |

# 3 Examples

## 3.1. Login to FTP Server

//Step 1: Configure and activate the PDP context.

**AT+QICSGP=1,1,"UNINET","", "",1** //Set PDP context as 1 and China Unicom APN as "UNINET".

OK

**AT+QIACT=1** //Activate PDP context 1.

OK //Activated successfully.

**AT+QIACT?** //Query the state of PDP context.

**+QIACT: 1,1,1,"10.7.157.1"**

OK

**AT+QFTPCFG="contextid",1** //Set the PDP context ID as 1. The PDP context ID must be activated first.

OK

//Step 2: Configure user account and transfer settings.

**AT+QFTPCFG="account","test","test"** //Set username and password.

OK

**AT+QFTPCFG="filetype",1** //Set file type as ASCII.

OK

**AT+QFTPCFG="transmode",1** //Set transfer mode as passive mode.

OK

**AT+QFTPCFG="rsptimeout",90** //Set the maximum response time as default value 90.

OK

//Step 3: Login to FTP server.

**AT+QFTPOPEN="220.180.239.212",8309**

OK

**+QFTPOPEN: 0,0**

## 3.2. Login to FTPS Server

//Step 1: Configure and activate the PDP context.

**AT+QICSGP=1,1,"UNINET","",1** //Set PDP context 1. China Unicom APN: "UNINET".

OK

**AT+QIACT=1** //Activate PDP context 1.

OK //Activated successfully.

**AT+QIACT?** //Query the state of PDP context.

**+QIACT: 1,1,1,"10.7.157.1"**

OK

**AT+QFTPCFG="contextid",1** //Set the PDP context ID as 1. The PDP context ID must be activated first.

OK

//Step 2: Configure user account and transfer settings.

**AT+QFTPCFG="account","test","test"** //Set username and password.

OK

**AT+QFTPCFG="filetype",1** //Set file type as ASCII.

OK

**AT+QFTPCFG="transmode",1** //Set transfer mode as passive mode.

OK

**AT+QFTPCFG="rsptimeout",90** //Set response timeout value.

OK

//Step 3: Configure FTPS.

**AT+QFTPCFG="ssltype",1** //Set SSL type as 1, the module is a FTPS client.

OK

**AT+QFTPCFG="sslctxid",1** //Set SSL context as 1.

OK

**AT+QSSLCFG="ciphersuite",1,0xffff** //Set SSL cipher suite type as 0xffff, which supports all cipher suite types.

OK

**AT+QSSLCFG="secllevel",1,0** //Set SSL security level as 0, which means the SSL CA certificate is not needed.

OK

**AT+QSSLCFG="sslversion",1,1** //Set SSL version as 1, which means TLS1.0.

OK

//Step 4: Login to FTPS server.

**AT+QFTPOPEN="quectel.3322.org",990**

OK

**+QFTPOPEN: 0,0**

### 3.3. Folder Operation

```

AT+QFTPCWD="/" //Set current directory.
OK

+QFTPCWD: 0,0
AT+QFTPPWD //Query current directory.
OK

+QFTPPWD: 0,/
AT+QFTPMKDIR="TEST" //Create a folder as TEST on FTP(S) server.
OK

+QFTPMKDIR: 0,0
AT+QFTPRENAME="TEST","TEST_NEW" //Rename the folder as TEST_NEW.
OK

+QFTPRENAME: 0,0
AT+QFTPRMDIR="TEST_NEW" //Delete the folder TEST_NEW.
OK

+QFTPRMDIR: 0,0

```

### 3.4. File Operation

```

AT+QFTPCWD="/" //Set current directory.
OK

+QFTPCWD: 0,0
AT+QFTPPWD //Query current directory.
OK

+QFTPPWD: 0,/"
AT+QFTPSIZE="test_my1.txt" //Query the size of test_my1.txt on FTP(S) server.
OK

+QFTPSIZE: 0,1000
AT+QFTPRENAME="test_my1.txt","test_new.txt" //Rename the file as test_new.txt.
OK

```

```

+QFTPFILENAME: 0,0
AT+QFTPMDTM="test_new.txt" //Get the file modification time of test_new.txt on FTP(S) server.
OK

+QFTPMDTM: 0,"20140708110039"
AT+QFTPDEL="test_new.txt" //Delete test_new.txt on FTP(S) server.
OK

+QFTPDEL: 0,0

```

### 3.5. List File Information or File Names

```

AT+QFTPCWD="/" //Set current directory.
OK

+QFTPCWD: 0,0
AT+QFTPLIST="." //List the content of current directory and the data are outputted to
COM port.

CONNECT
<Output content data>
OK

+QFTPLIST: 0,1000
AT+QFTPLIST="TEST_2","COM:" //List the content of TEST_2 and the data are outputted to COM port.
CONNECT
<Output content data>
OK

+QFTPLIST: 0,1000
AT+QFTPNLST="." //List file names of current directory and the data are outputted to
COM port.

CONNECT
<Output content data>
OK

+QFTPNLST: 0,1000
AT+QFTPNLST="TEST_2","COM:" //List file names of TEST_2 and the data are outputted via COM
port.

CONNECT
<Output content data>
OK

```

```
+QFTPNLST: 0,1000
AT+QFTPMLSD="." //List standardized file and directory information of current directory
and the data are outputted via COM port.

CONNECT
<Output content data>
OK

+QFTPMLSD: 0,1000
AT+QFTPMLSD="TEST_2","COM:" //List standardized directory information of TEST_2 and the data
are outputted via COM port.

CONNECT
<Output content data>
OK

+QFTPMLSD: 0,1000
```

### 3.6. Upload a File to FTP(S) Server

```
AT+QFTPCWD="/"
OK

+QFTPCWD: 0,0
AT+QFTPSTAT
+QFTPSTAT: 0,1

OK

//Upload a file via COM port.

AT+QFTPPUT="test_my1.txt","COM:",0 //All data will be saved as test_my1.txt on FTP(S) server.
CONNECT
<Input file data>
<++++>
OK

+QFTPPUT: 0,1000
AT+QFTPLEN
OK

+QFTPLEN: 0,1000
AT+QFTPSIZE="test_my1.txt"
OK
```

```
+QFTPSIZE: 0,1000
```

```
//Upload a file via COM port and the start position is 1000.
```

```
AT+QFTPPUT="test_my1.txt","COM:",1000 //All data will be saved as test_my1.txt on FTP(S) server.
```

```
CONNECT
```

```
<Input file data>
```

```
<++++>
```

```
OK
```

```
+QFTPPUT: 0,500
```

```
AT+QFTPSIZE="test_my1.txt"
```

```
OK
```

```
+QFTPSIZE: 0,1500
```

```
//Solution 1: Upload a file via COM port to FTP(S) server twice in 1024-byte chunks .
```

```
AT+QFTPPUT="test_my1.txt","COM:",0,1024,0 //It is not the last 1024-byte chunk of test_my1.txt.
```

```
CONNECT
```

```
<Input file data>
```

```
OK
```

```
//Data length reaches 1024 bytes.
```

```
+QFTPPUT: 0,1024
```

```
AT+QFTPPUT="test_my1.txt","COM:",1024,1024,1 //It is the last 1024-byte chunk of test_my1.txt.
```

```
CONNECT
```

```
<Input file data>
```

```
OK
```

```
//Data length reaches 1024 bytes.
```

```
+QFTPPUT: 0,1024
```

```
//Solution 2: Upload a file from UFS to FTP(S) server.
```

```
AT+QFUPL="UFS:test_ufs.txt",1000,300,1 //Upload a file to UFS, the file will be saved as test_ufs.txt
and the maximum file size is 1000 bytes. 300 indicates
timeout and 1 indicates ACK mode. For more details, see
document [4].
```

```
CONNECT
```

```
<Input 1000 bytes data>
```

```
+QFUPL: 1000,707
```

```
OK
```

```
AT+QFLST="UFS:/"
```

```
+QFLST: "UFS:test_ufs.txt",1000
```

```
OK
```

```
AT+QFTPPUT="test_my1.txt","UFS:test_ufs.txt",0 //Upload UFS:test_ufs.txt to FTP(S) server and
```

save as *test\_my1.txt* on FTP(S) server.

OK

+QFTPPUT: 0,1000

//Solution 3: Upload a file from SD to FTP(S) server.

**AT+QFUPL="SD:test\_sd.txt",1000,300,1**

//Upload a file to SD, the file is saved as *test\_sd.txt* and the maximum file size is 1000 bytes. 300 indicates timeout and 1 indicates ACK mode. For more details, see **document [4]**.

CONNECT

<Input 1000 bytes data>

+QFUPL: 1000,707

OK

**AT+QFLST=SD:\***

+QFLST: "SD:test\_sd.txt",1000

OK

**AT+QFTPPUT="test\_my1.txt","SD:test\_sd.txt",0**

//Upload *SD:test\_sd.txt* to FTP(S) server and save as *test\_my1.txt* on FTP(S) server.

OK

+QFTPPUT: 0,1000

**AT+QFTPLEN**

OK

+QFTPLEN: 0,1000

**AT+QFTPSIZE="test\_my1.txt"**

OK

+QFTPSIZE: 0,1000

### 3.7. Download a File from FTP(S) Server

```

AT+QFTPCWD="/"
OK

+QFTPCWD: 0,0

//Solution 1: Output downloaded data directly via COM port.

//Download a file from FTP(S) server and the data are outputted via COM port.

AT+QFTPGET="test_my.txt","COM:"
CONNECT
<Output file data>
OK

+QFTPGET: 0,1000

//Download a file. The data will be outputted via COM port twice in 500-byte chunks.

AT+QFTPGET="test.txt","COM:",0,500           //The size of test.txt is 1000 bytes. Download the
                                              first 500 bytes.

CONNECT
<Output file data>
OK

+QFTPGET: 0,500
AT+QFTPGET="test.txt","COM:",500,500         //Download the remaining 500 bytes.
CONNECT
<Output file data>
OK

+QFTPGET: 0,500

//Solution 2: Save downloaded data to UFS file.

//Download a file from FTP(S) server and save it to UFS.

AT+QFTPGET="test_my1.txt","UFS:test.txt"     //Download the file and save it to UFS as test.txt.
OK

+QFTPGET: 0,1000
AT+QFLST="UFS:"
+QFLST: UFS:test.txt,1000

OK

```

//Download a file from FTP(S) server and save it to UFS, the start position is 450.

**AT+QFTPGET="test\_my1.txt","UFS:test1.txt",450** //Download file and save it to UFS as *test1.txt*.  
OK

**+QFTPGET: 0,550**

//Solution 3: Save downloaded data to SD file.

//Download a file from FTP(S) server and save it to SD.

**AT+QFTPGET="test\_my1.txt","SD:test.txt"** //Download a file and save it to SD card as *test.txt*.  
OK

**+QFTPGET: 0,1000**

**AT+QFLST="SD:\***

**+QFLST: SD:test.txt,1000**

OK

//Download a file from FTP(S) server and save it to SD. The start position is 450.

**AT+QFTPGET="test\_my1.txt","SD:test1.txt",450** //Download file and save it to SD card as *test.txt*.  
OK

**+QFTPGET: 0,550**

**AT+QFTPLEN**

OK

**+QFTPLEN: 0,550**

**AT+QFLST="SD:\***

**+QFLST: SD:test.txt,1000**

**+QFLST: SD:test1.txt,550**

OK

### 3.8. Log out from FTP(S) Server

**AT+QFTPCLOSE** //Log out from FTP(S) server.

OK

**+QFTPCLOSE: 0,0**

**AT+QIDEACT=1** //Deactivate the PDP context, which was activated for FTP(S).

OK

# 4 Error Handling

## 4.1. Executing FTP(S) AT Command Failure

If **ERROR** response is received from the module when executing FTP(S) AT commands, check whether:

- the (U)SIM card is inserted.
- **+CPIN: READY** is returned when executing **AT+CPIN?**.

## 4.2. PDP Activation Failure

If a PDP context activation with **AT+QIACT** fails, check the following configurations:

1. Query whether the PS domain is attached or not with **AT+CGATT?**, if not, please execute **AT+CGATT=1** to attach the PS domain.
2. Query the PS domain status with **AT+CGREG?** and make sure the PS domain has been registered.
3. Query the PDP context parameters with **AT+QICSGP** and make sure the APN of the specified PDP context has been set.
4. Make sure the specified PDP context ID is neither used by PPP nor activated via **AT+CGACT**.
5. According to 3GPP specifications, the module only supports three simultaneously activated PDP contexts, therefore make sure the number of activated PDP contexts is no more than 3.

If all above configurations are correct, but activating the PDP context with **AT+QIACT** still fails, reboot the module to resolve this issue. After rebooting the module, check the configurations above at least three times in 10-minute intervals to avoid frequent module rebooting.

### 4.3. DNS Parsing Failure

When executing **AT+QFTPOPEN**, if **+QFTPOPEN: 604,0** is returned, please check the following:

1. Make sure the domain name of FTP(S) server is valid.
2. Query the status of the PDP context with **AT+QIACT?** to make sure the specified PDP context has been activated successfully.

### 4.4. Error Response of FTP(S) Server

If the **<protocol\_error>** in **+QFTPXX: <err>,<protocol\_error>** is not 0, it indicates an error code from FTP(S) server.

The issue can be identified based on the returned protocol error code. For example, if **<protocol\_error>** is 530 (not logged in), it indicates that **<username>** or **<password>** may be wrong. If **<protocol\_error>** is 550 (requested action not taken: file unavailable), it means that the file or directory may not exist. For more details, see *RFC959*.

# 5 Summary of Error Codes

The detailed descriptions of **<err>** are provided in the following table.

**Table 3: Summary of Error Codes**

| <b>&lt;err&gt;</b> | <b>Description of Error Codes</b>                 |
|--------------------|---------------------------------------------------|
| 0                  | Operation successful                              |
| 601                | Unknown error                                     |
| 602                | FTP(S) server blocked                             |
| 603                | FTP(S) server busy                                |
| 604                | DNS parse failed                                  |
| 605                | Network error                                     |
| 606                | Control connection closed.                        |
| 607                | Data connection closed                            |
| 608                | Socket closed by peer                             |
| 609                | Timeout error                                     |
| 610                | Invalid parameter                                 |
| 611                | Failed to open file                               |
| 612                | File position invalid                             |
| 613                | File error                                        |
| 614                | Service not available, closing control connection |
| 615                | Open data connection failed                       |
| 616                | Connection closed; transfer aborted               |
| 617                | Requested file action not taken                   |

---

|     |                                                         |
|-----|---------------------------------------------------------|
| 618 | Requested action aborted: local error in processing     |
| 619 | Requested action not taken: insufficient system storage |
| 620 | Syntax error, command unrecognized                      |
| 621 | Syntax error in parameters or arguments                 |
| 622 | Command not implemented                                 |
| 623 | Bad sequence of commands                                |
| 624 | Command parameter not implemented                       |
| 625 | Not logged in                                           |
| 626 | Need account for storing files                          |
| 627 | Requested action not taken                              |
| 628 | Requested action aborted: page type unknown             |
| 629 | Requested file action aborted                           |
| 631 | SSL authentication failed                               |

---

# 6 Summary of FTP(S) Protocol Error Codes

The protocol error code **<protocol\_error>** indicates an error response from FTP(S) server. See *RFC959* for more information. Some common **<protocol\_error>** are described in the following table.

**Table 4: Summary of FTP(S) Protocol Error Codes**

| <b>&lt;protocol_error&gt;</b> | <b>Meaning</b>                                             |
|-------------------------------|------------------------------------------------------------|
| 421                           | Service not available, closing control connection          |
| 425                           | Open data connection failed                                |
| 426                           | Connection closed; transfer aborted                        |
| 450                           | Requested file action not taken                            |
| 451                           | Requested action aborted: local error in processing        |
| 452                           | Requested action not taken: insufficient system storage    |
| 500                           | Syntax error, command unrecognized                         |
| 501                           | Syntax error in parameters or arguments                    |
| 502                           | Command not implemented                                    |
| 503                           | Bad sequence of commands                                   |
| 504                           | Command parameter not implemented                          |
| 530                           | Not logged in                                              |
| 532                           | Need account for storing files                             |
| 550                           | Requested action not taken: file unavailable               |
| 551                           | Requested action aborted: page type unknown                |
| 552                           | Requested file action aborted: exceeded storage allocation |
| 553                           | Requested action not taken: file name not allowed          |

# 7 Appendix References

**Table 5: Related Documents**

| Document Name                                                           |
|-------------------------------------------------------------------------|
| [1] Quectel_ECx00K&ECx00M&ECx00N&EG810M_Series_TCP(IP)_Application_Note |
| [2] Quectel_ECx00K&ECx00M&ECx00N&EG810M_Series_AT_Commands_Manual       |
| [3] Quectel_ECx00K&ECx00M&ECx00N&EG810M_Series_SSL_Application_Note     |
| [4] Quectel_ECx00K&ECx00M&ECx00N&EG810M_Series_FILE_Application_Note    |

**Table 6: Terms and Abbreviations**

| Abbreviation | Description                                        |
|--------------|----------------------------------------------------|
| ACK          | Acknowledgement                                    |
| APN          | Access Point Name                                  |
| ASCII        | American Standard Code for Information Interchange |
| DNS          | Domain Name Server                                 |
| DTR          | Data Terminal Ready                                |
| FTP          | File Transfer Protocol                             |
| FTPS         | FTP over SSL                                       |
| ID           | Internet Protocol                                  |
| PDP          | Packet Data Protocol                               |
| LTE          | Long-Term Evolution                                |
| PDP          | Packet Data Protocol                               |
| PPP          | Point-to-Point Protocol                            |

|        |                                        |
|--------|----------------------------------------|
| PS     | Packet Switching                       |
| QoS    | Quality of Service                     |
| RAM    | Random Access Memory                   |
| SD     | Secure Digital                         |
| SSL    | Secure Sockets Layer                   |
| TLS    | Transport Layer Security               |
| UFS    | Universal Flash Storage                |
| URC    | Unsolicited Result Code                |
| (U)SIM | (Universal) Subscriber Identity Module |
| WM     | Watermark                              |